

Pratyush Raj Gupta

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EDUCATION

Texas A&M University

Bachelor of Science in Computer Science, Minor in Statistics

College Station, TX

Aug. 2022 – May 2026

EXPERIENCE

ML Developer

Polara Labs

March 2026 – Present

College Station, TX

- Engineered an **AI-driven FastAPI/Docker microservice** that evaluates **SOC 2 audit evidence**, using **semantic matching**, **hybrid scoring**, and **gated LLM adjudication** to automate audit review.
- Delivered **100% accuracy** on **88 test artifacts** with **sub-3s latency**, enabling real-time feedback and cutting costs

Software Engineering Intern

Region 10 ESC

May 2025 – Aug. 2025

Richardson, TX

- Architected a **RAG-powered natural language-to-SQL interface** that converted business questions into production-ready queries and results, reducing data request turnaround by **50%** for **non-technical users**.
- Engineered a **multi-cloud data lakehouse architecture** using **Apache Iceberg**, automating **Snowflake** exports directly to **AWS S3** and integrating with **Oracle Autonomous Database**.

Software Engineering Intern

Region 10 ESC

May 2024 – Aug. 2024

Richardson, TX

- Created an **API-to-Oracle Cloud ETL pipeline** using **PL/SQL**, reducing processing time by **70%**.
- Built a **full-stack attendance app** serving **10+ school districts** using a **PL/SQL logic engine** for funding eligibility.
- Engineered an **Oracle APEX funding projection dashboard** using **JavaScript** and **SQL** for **10+ school districts**.

Undergraduate Researcher

Texas A&M Texas Water Observatory

Aug. 2023 – Dec. 2023

College Station, TX

- Architected a **Python/PostgreSQL pipeline** replacing manual CSV workflows, increasing processing throughput by **95%**.
- Implemented automated integrity validations and real-time outlier alerting for high-fidelity telemetry reporting.

PROJECTS

Hybrid Phishing Detector | Python, PyTorch, XGBoost, DistilBERT, BiLSTM, SHAP/LIME

Mar. 2026 – May 2026

- Trained an **XGBoost + DistilBERT/BiLSTM ensemble** on **149K emails** for 3-class phishing detection.
- Achieved **98.55% accuracy** and **99.87% AUC-ROC** with **SHAP/LIME** interpretability.

Market Sentinel | Java, Spring Boot, PostgreSQL, Redis, Docker, REST APIs

Dec. 2025 – Present

- Architected a **Java/Spring Boot** market monitoring REST API to ingest macroeconomic data from multiple providers.
- Implemented **Redis caching** and **Docker containerization**, reducing latency and optimizing compute resources.
- Designed a data abstraction layer with **idempotent writes** to ensure **100% consistency** during API throttling.

A&M Grade Data Insights | Python, FastAPI, pgvector, React, AWS RDS (MySQL)

Dec. 2024 – Jan. 2025

- Built a **Python/FastAPI RAG conversational agent** with **pgvector** to analyze academic data via natural language.
- Engineered a **Python ETL pipeline** to scrape and normalize **10,000+ records** into a high-availability **AWS RDS** instance.
- Developed and deployed a **React frontend** to visualize complex database insights for data-backed course selection.

TAMU Legion Web Platform | Ruby on Rails, PostgreSQL, Docker, GitHub Actions

Aug. 2025 – Dec. 2025

- Led **Agile development** as **Project Manager** of a **Ruby on Rails SaaS portal**, centralizing operations with strict **RBAC**.
- Built **Docker/GitHub Actions CI/CD pipelines** enforcing **unit tests with 90%+ code coverage** and linting

LEADERSHIP

President

Beta Tau Omega Engineering Fraternity Inc.

Aug. 2024 – May 2025

College Station, TX

- Spearheaded **40% membership growth** and managed a **\$10K+ annual operating budget**.
- Directed philanthropy and technical marketing initiatives, earning recognition in the **Gastric Cancer Foundation magazine**.

TECHNICAL SKILLS

Languages: Python, Java, C++, SQL, PL/SQL, JavaScript, TypeScript, C#, Swift, Ruby

Backend & Cloud: AWS (S3, RDS, Lambda, EC2), GCP, Oracle Cloud, Docker, Kubernetes, Spring Boot, Node.js, REST APIs

Data & AI: Snowflake, Apache Iceberg, Spark, PostgreSQL, MySQL, Redis, pgvector, PyTorch, Pandas, RAG pipeline, NumPy

Tools & DevOps: Git, GitHub Actions, CI/CD, Terraform, Linux, Agile/Scrum, React, Jira